

India @ AI & Deeptech Crossroads

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As SpaceX, OpenAI, and Anthropic get listed at astronomical valuations or start on a roadshow to enter public markets with a trillion dollar tag, the mood in India show is mixed. Given the excitement around these, we could call the coming decade the decade of AI and deeptech, exemplified by such companies that seek to dominate their respective fields not only on Earth, but in space and other domains too. Public markets from India on the other hand, have been underperformers, in the time when AI and deeptech sectors have at least quadrupled in value in just a few years. The reason so far has been India has neither the frontier AI models and private space sector of the US, nor the massive manufacturing powerhouses of East Asia. Yet for India, that change may be closer than it looks, and the pieces to turn it around are already coming together. The biggest piece is the frugal innovation, both for ai and deeptech including space compute. India stands at a crossroads, with low AI adoption on business side at the point, but that's likely to change fast

India has a clear shot to be the world's capital of the AI app layer, which will be worth an estimated USD 50 trillion in the next decade. Deeptech is likewise going through a huge metamorphosis, with scrappy young entrepreneurs working hard to lead the world by leveraging a huge domestic market. The driver in both the cases will be innovation driven by frugal mindset, both on land and space, both in ai and deeptech.

Clearly then, India needs, and is luckily in the throes of, a renaissance of its own, where the ancient knowledge culture of the Saraswati civilisation potentially merges with the future epoch of AI and deeptech. But some critical pieces need to come together for that vision.

The Diagnosis: Success That Softened Us

First, let's diagnose why we are here, far lower in the global pecking order as of now. Paradoxically, it was the success of the services economy that built the Indian growth machine this century which also softened the innovation and risk taking capacity of the Indian tech sector.. 5–7 lakh crore (~USD 600B) returned by Indian IT services to investors in the last decade against just USD 10–20B invested in Indian AI & deeptech in the same period for liberal ‘AI washing.’

As veterans like Uday Kotak have pointed out, while Google and other highly profitable tech companies have been investing more, year on year, into AI and deeptech, assuming they will compound capital, profitable companies out of India, whether from tech or other sectors, have been happy returning capital to shareholders to juice up equity returns rather than get into knowledge heavy AI and deeptech sectors.

A deeper issue is the lack of confidence in the future direction of the world and lack of risk appetite to take bold bets to shape the future, given the disconnected ecosystem. While no one, including Nobel laureates or the most successful AI lab founders, can accurately forecast the trajectory of AI in decades, years, or even months from now, everyone can see the magic unfolding with the rapidly improving capabilities of frontier AI models. While this is yet to show up in tangible returns on investment for the whole economy in the form of much higher GDP, that may be too soon to expect for a transformation as epochal as this. So despite persistent talk of an AI bubble, investment continues in the US and China, the twin engines of global growth, leaving India far behind as of now. The bean counters at the helm of most corporates, large or startup, have chosen to stay away from rollercoaster of creation and innovation, and are happy to apply a safe return on investment metric which is mostly like driving into a storm while looking in the rearview mirror instead of ahead. While India produces one of the highest numbers of engineers in the world, most of them have been happily doing corporate jobs, writing code, or becoming program managers all of which today are marked by a lack of innovation, passion, or the zeal to be the best in the world.

The Silver Lining: A Renaissance in the Making

However, as they say, every cloud has a silver lining. This is true for India today, and has always been. The next generation of entrepreneurs, luckily, carries no baggage of scarcity or a narrow services mindset, and is willing to dream big. This strengthens the case for rapid growth in the AI app layer, where India could leverage its massive talent to build solutions for a multilingual, culturally diverse country that is more like a continent. In deeptech, India is slowly and steadily making its mark, from rocket propulsion for reusable rockets to autonomy software for drones, quantum, and chips, bringing together academic rigour with a diehard spirit of continuous experimentation and improvement. Solar

and hydrogen revolutions offer glimpses of the future of energy, and nascent activity in green data centres shows the promise of a new sovereign stack. The biggest thing in favour of India is ultimately the frugal mindset, which makes it feasible to do transformational work while spending a fraction of what is spent by equivalent in US or Europe.

Best example of this frugal innovation is space, where the historic Chandrayaan-3 mission by ISRO, which successfully landed a rover on the lunar south pole, cost about \$75 million, while NASA's comparable planetary missions and proposed lunar lander contracts typically exceed \$600 million to \$2.7 billion

The stark contrast in budgets stems from India's frugal engineering models and mindsets Cost Breakdown. The app layer in AI is exactly the same story, as a scrappy founder in India's Nashik or Varanasi has access to the best of frontier models of US or open source models of China at the same time and same cost as anyone else, and can build AI applications far more cheaply than almost anyone else in the world, as was shown by the success of the IT industry.

Government policy also helps, with large new funds coming into play to help startups fight the valley of death in the long journey from idea to lab to market. And the new crop of young entrepreneurs is risk taking, willing to sacrifice, and working both hard and smart. What is needed is a Silicon Valley like ecosystem, one that brings corporates and startups, as profit seeking enterprises, into the mix with nonprofit government and academia, for the greater good of all. Perhaps the missing piece is the glue that brings this holy trinity of nonprofit and for profit together like the invisible Saraswati at the holy Sangam of Prayagraj.

The healthy economic growth of the last few decades has created enough, and more, private capital in India, thanks to the transformational growth of financial services across mutual funds and insurance. That pool of capital now needs to be channelised for the true growth of the country in AI and deeptech, and then the foreign capital will flow through seeking best execution at lowest cost. Deliberate architecture and design of systems to support the markets could be the invisible glue that catalyses innovation, boosts investment in the fledgling AI and deeptech sectors, and creates the ecosystem needed to see at least one Indian company, or fund, reach the vaunted perch of a trillion dollars in the next decade.

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